Incident Report Analysis

Summary	Earlier today, our company experienced a DoS attack that compromised our internal networks resulting in having to shut down the internal network for two hours until the issue was resolved. The analyst team noticed a flood of ICMP packets to the internal networks suspending activity to resolve the ICMP packets. To resolve the issue, the team suspended blocked all ICMP packets to restore critical network operations before resuming operations. After restoring network access, the security team bolstered the firewall security and imposed stricter access permissions for incoming traffic along with implementing identifying and preventative measures for abnormal traffic.
Identify	The security team was able to identify an unconfigured firewall in discovery. This allowed for the attacker to flood the network with ICMP pings through an essential backdoor.
Protect	The security team has implemented two countermeasures in light of the DoS attack to prevent future attacks. The first countermeasure was to implement a new firewall rule to implement the rate of incoming ICMP packets. This will immediately reduce the impact by stopping a request that could potentially overload the internal network. The second countermeasure was to implement a source IP address verification on the firewall. This will allow for the firewall system to check for IP spoofing on incoming ICMP packets and ensure that questionable IP addresses cannot gain access to the network. Network monitoring software to detect abnormal traffic patterns
Detect	To detect any unwanted traffic in the future, the security team has

	implemented a networking monitoring software to detect abnormal traffic patterns.
Respond	By using IDS/IPS systems to filter out some ICMP traffic based on suspicious characteristics, the security team will be able to proactively prevent these issues from arising.
Recover	Security teams will leverage IDS/IDP data and use baseline image to restore data.

Reflections/Notes: This incident reinforced the importance of being proactive in network security by continuous monitoring of systems.
The DoS attack demonstrated how an overlooked firewall configuration can create a significant vulnerability and disrupt critical network operations. While blocking ICMP traffic restored access, the response also showed the root cause of the attack and ways to prevent future attacks. Implementing ICMP rate limiting, source IP verification, and IDS/IPS monitoring provides multiple layers of protection against abnormal traffic. Overall, this scenario demonstrated that cybersecurity requires both effective incident response and preventative measures to identify vulnerabilities before they can be exploited.